PRODUCT USER MANUAL

For
OMI_CLIMATE_OHC_BALTIC_area_averaged_anomalies

Issue: 2.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_OHC_BALTIC_area_averaged_anomalies. It contains one dataset:

- omi_climate_ohc_baltic_area_averaged_anomalies

The OMI provides a timeseries of Baltic Sea area averaged ocean heat content anomalies. The method for calculating the ocean heat content anomaly is based on the daily mean sea water potential temperature fields (Tp) derived from the Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The total heat content is determined using the following formula:

$HC = \rho * cp * (Tp + 273.15)$.

Here, ρ and cp represent spatially varying sea water density and specific heat, respectively, which are computed based on potential temperature, salinity and pressure using the UNESCO 1983 polynomial developed by Fofonoff and Millard (1983).

The vertical integral is computed using the static cell vertical thicknesses sourced from the reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 dataset cmems_mod_bal_phy_my_static, spanning from the sea surface to the ocean bottom. Spatial averaging is performed over the Baltic Sea spatial domain, defined as the region between 13° - 31° E and 53° - 66° N.

To obtain the annual anomaly timeseries in (J/m2), the annual mean total heat content is subtracted from the mean heat content over the reference period of 1993-2014.

We evaluate the uncertainty from the mean annual error of the potential temperature compared to the observations from the Baltic Sea (Giorgetti et al., 2020). The shade corresponds to the RMSD of the annual mean heat content biases (± 35.3 MJ/m²) evaluated from the observed temperatures and corresponding model values.

The the linear trends (W/m2) have been evaluated from the annual anomalies with the uncertainty of 1.96* standard error.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_OHC_BALTIC_area_averaged_anomalies
Geographical coverage	Baltic Sea
Variables	Ocean heat content anomaly
Available time series	From 1993 to Year-1
Temporal resolution	Yearly
Horizontal resolution	Integrated
Integration depth (for example)	0 – 300 m
Format	NetCDF4

b) Production System Description

The Baltic Sea area averaged ocean heat content anomalies have been calculated using the results from Copernicus Marine Service BALMFC Reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The vertical integral is computed using the static cell vertical thicknesses sourced from the reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 dataset cmems_mod_bal_phy_my_static, spanning from the sea surface to the ocean bottom. Spatial averaging is performed over the Baltic Sea spatial domain, defined as the region between 13° - 31° E and 53° - 66° N.

c) Dataset details

omi_climate_ohc_baltic_area_averaged_anomalies contains the following variables:
ohc_balrean [J m^{-2}] long_name: Area Averaged Ocean Heat Content Anomaly from Baltic Sea Physical Reanalysis standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content **ohc_balrean_lower_rmsd** [J m^{-2}] long_name: Lower Bound of Error of the Area Averaged Ocean Heat Content Anomaly from Baltic Sea Physical Reanalysis standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content **ohc_balrean_upper_rmsd** [J m^{-2}] long_name: Upper Bound of Error of the Area Averaged Ocean Heat Content Anomaly from Baltic Sea Physical Reanalysis standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content **time** [days since 1950-01-01 00:00:00] long_name: time standard_name: time

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-CLIMATE-OHC-area-averaged-anomalies)
Fofonoff, P. and Millard, R.C. Jr UNESCO 1983. Algorithms for computation of fundamental properties of seawater. UNESCO Tech. Pap. in Mar. Sci., No. 44, 53 pp., p.39. http://unesdoc.unesco.org/images/0005/000598/059832eb.pdf

Giorgetti, A., Lipizer, M., Molina Jack, M.E., Holdsworth, N., Jensen, H.M., Buga, L., Sarbu, G., Iona, A., Gatti, J., Larsen, M. and Fyrberg, L., 2020. Aggregated and Validated Datasets for the European Seas: The Contribution of EMODnet Chemistry. Frontiers in Marine Science, 7, p.583657.